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Interpretable and explainable predictive machine learning models for gastric retention before gastroscopy

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Background: Early detection of gastric retention (GR) is challenging in clinical practice. We aimed to develop and validate interpretable machine learning (ML) models for predicting GR risk using routinely available clinical features.

Methods: This retrospective study enrolled 3,233 consecutive patients (training cohort: $n = 2,263$; validation cohort: $n = 970$) who underwent gastroscopy between January 2022 and December 2024. To address class imbalance, the Synthetic Minority Over-sampling Technique (SMOTE) was applied prior to model development. Optimal features were selected via least absolute shrinkage and selection operator (LASSO) regression within the SMOTE-enhanced training cohort, with different predictive models constructed using nine machine learning (ML) algorithms. Ten-fold cross-validation on SMOTE-enhanced training cohort was used for hyperparameter tuning. Model performance was primarily assessed on the validation cohort using the area under the receiver operating characteristic curve (AUC), calibration plots, and decision curve analysis (DCA).

Results: LASSO regression identified 12 clinically actionable predictors from 19 candidate variables, which were subsequently used to develop nine ML models. Among these models, the extreme gradient boosting (XGBoost) model achieved the highest AUC value of 0.710 (95% CI: 0.681–0.738) in the validation cohort. Decision curve analysis revealed that the XGBoost model exhibited superior clinical utility compared to other models. Shapley Additive Explanations (SHAP) identified the use of anticholinergic drugs, pyloric obstruction, analgesic use, systemic diseases, and a history of gastrointestinal surgery as the top five important predictive features.

Conclusions: This study establishes an interpretable GR risk stratification model using routinely available preoperative variables. The model may serve as an auxiliary screening aid for pre-gastroscopy risk assessment.

KEYWORDS

gastric retention, gastroscopy, machine learning, predictive model, risk stratification

Introduction

Gastric retention (GR) is a clinically significant condition characterized by delayed food emptying and diagnosed during gastroscopy (1). Although fasting for 6–8 h is routine before gastroscopy, it does not seem to be effective for some patients. Previous studies have shown that patients with diabetes and a history of gastrointestinal surgery

are prone to GR (2–5). A systematic study indicated that GR can be detected in approximately 40% of patients with type 1 diabetes and 20% of patients with type 2 diabetes (6). Patients with GR face a high risk of aspiration, which can lead to pneumonia or even asphyxiation, especially when undergoing anesthetized gastroscopy. Moreover, traditional risk stratification methods lack precision in identifying high-risk individuals, particularly in populations with heterogeneous metabolic profiles or polypharmacy use. With the global increase in diabetes, obesity, and surgical interventions, there is an urgent need for predictive tools that integrate multifactorial clinical data to enable early risk stratification and personalized management.

Recent advances in machine learning (ML) have demonstrated transformative potential in clinical prediction tasks. For instance, interpretable ML models have successfully predicted conditions such as extended-spectrum β -lactamase (ESBL)-positive urinary tract infections and tuberculosis development by leveraging routine clinical variables, achieving high accuracy (AUC >0.90) and clinical interpretability (7, 8). Notably, ensemble algorithms like XGBoost have emerged as robust frameworks for handling imbalanced datasets and capturing nonlinear interactions among predictors, as evidenced by their superior performance in predicting cardiovascular drug use and metabolic risk trajectories (9, 10). A key challenge in developing ML models for GR lies in balancing predictive performance with clinical interpretability. Recent studies highlight the value of explainable AI techniques, such as Shapley Additive Explanations (SHAP), in elucidating feature contributions and guiding actionable interventions (11, 12). For example, SHAP analysis in predicting Parkinson's disease and surgery-related cardiovascular events revealed modifiable risk factors, such as medication use and biomarker thresholds, which informed targeted therapeutic strategies (13–15).

However, the application of ML to GR prediction remains underexplored. Aspiration caused by gastric retention can lead to severe pneumonia or even asphyxiation, which exacerbates the disease burden on patients and the social medical burden resulting from repeated examinations. Herein, we retrospectively analyzed a large number of patients with GR to develop and validate an interpretable ML model for GR risk prediction using routinely collected clinical data. Information obtained from this large-scale study aligns with emerging trends in precision medicine, where ML-powered tools are increasingly deployed to optimize diagnostic workflows and reduce complications in high-risk populations.

Materials and methods

Patients

This retrospective cohort study adhered to the ethical principles outlined in the 1964 Declaration of Helsinki and was approved by the Institutional Review Board of the First Affiliated Hospital of Xiamen University (XMY-2025KY173). Due to the retrospective nature of the study, written informed consent was abandoned by the ethics committee of the First Affiliated Hospital of Xiamen University. The study adheres to STROBE guidelines. GR was defined as having at least a 6.3% residual volume at four hours or the observation of unexpected

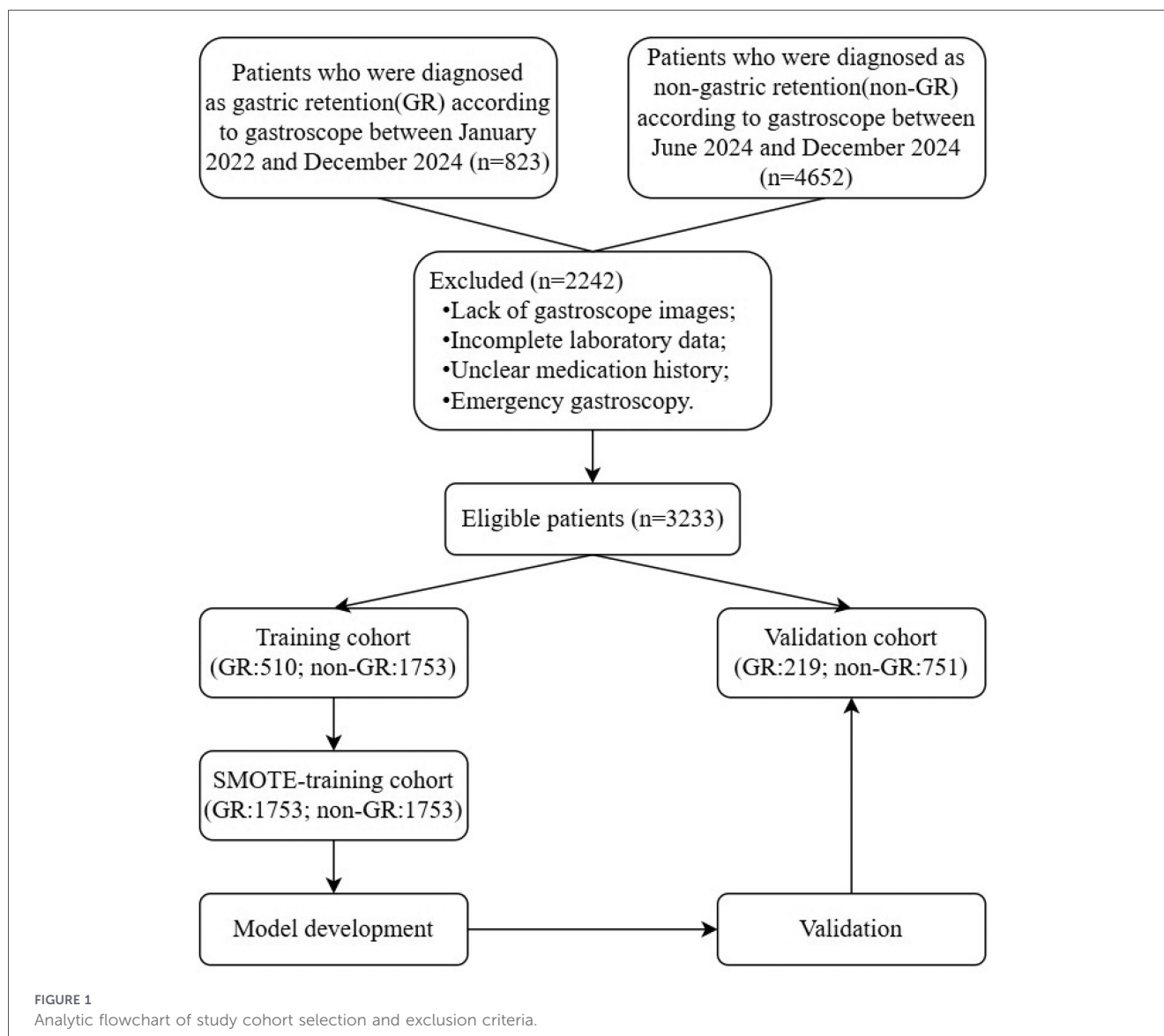
food residue after an 8 h fasting protocol during gastroscopy (1, 16). We searched the database of the First Affiliated Hospital of Xiamen University and identified patients diagnosed with GR via gastroscopy between January 2022 and December 2024. Non-GR patients who underwent gastroscopy between June 2024 and December 2024 were also included. All patients undergoing gastroscopy were prepared according to the routine fasting protocol. Patients were excluded based on the following criteria: (1) absence of gastroscopy images, (2) incomplete laboratory data, (3) unclear medication history, and (4) emergency gastroscopy. Consequently, a total of 3,233 patients were ultimately enrolled, consisting of 729 GR patients and 2,504 non-GR patients. Initial clinical information for each patient was extracted from medical records that were documented prior to the index gastroscopy, including basic demographic data (age, gender, education level), surgical history, laboratory data, and medication history. Importantly, predictor variables such as pyloric obstruction, intestinal obstruction, and systemic diseases were ascertained from pre-existing imaging reports, clinical diagnoses, or prior endoscopic records, rather than from the findings of the current index gastroscopy. The study population was randomly divided into a training cohort ($n = 2,263$, with 1,753 non-GR cases and 510 GR cases) and a validation cohort ($n = 970$, with 751 non-GR cases and 219 GR cases) at a ratio of 7:3 via random sampling. The analytic flowchart of the study cohort is illustrated in Figure 1.

Oversampling

To mitigate class imbalance in the training cohort, we applied the Synthetic Minority Over-sampling Technique (SMOTE) with hyperparameters set to `sampling_strategy = "auto"`, `k_neighbors = 5`, and `random_state = 42`. This method synthesizes new instances that closely emulate the underlying distribution of the minority class. Thereby, a balanced dataset was established which termed the SMOTE-enhanced training cohort, comprising both authentic and algorithmically generated samples (17). The validation cohort was not subjected to SMOTE and remained in its original imbalanced distribution for unbiased final evaluation.

Feature selection, model development, and validation

To identify clinically relevant predictors within the SMOTE-enhanced training cohort ($n = 3,506$), we performed least absolute shrinkage and selection operator (LASSO) regression after SMOTE oversampling, with the penalty parameter optimized via 10-fold cross-validation. Variance inflation factor (VIF) and pairwise Pearson correlation matrix were calculated for LASSO-selected predictors to evaluate multicollinearity; a VIF threshold of 5 was adopted to define potential multicollinearity. The selected features were subsequently used to develop predictive models using nine distinct machine learning approaches: logistic regression (LR), decision tree (DT), random forest (RF), support vector machine (SVM), extreme gradient boosting (XGBoost), light gradient boosting machine (LightGBM), K-nearest neighbors (KNN), naive Bayes (NB), and



artificial neural network (ANN). Hyperparameters for each algorithm were determined via grid search with 10-fold cross-validation on the SMOTE-enhanced training cohort. The final hyperparameter settings are listed in [Supplementary Table S1](#). All models were evaluated in the validation cohort.

The predictive performance of each model was thoroughly assessed using metrics relevant to clinical discrimination and calibration, including the area under the receiver operating characteristic curve (AUC), accuracy, sensitivity, specificity, positive predictive value (PPV), negative predictive value (NPV), confusion matrix analysis, calibration curves, and decision curve analysis (DCA). The optimal probability threshold for the best-performing model was determined using the Youden index. Statistical comparisons of AUC values were conducted using the DeLong test. To ensure robustness, the optimal model underwent further internal validation via 10-fold cross-validation within the SMOTE-enhanced training cohort. Finally, feature importance was interpreted and model outputs were visualized using SHapley Additive exPlanations (SHAP) to enhance clinical interpretability.

Statistical analysis

Statistical calculations were performed using multiple software packages, including IBM SPSS Statistics (Version 22.0; <https://www.ibm.com/spss>), R (Version 4.4.1; <https://www.r-project.org>), and Python (Version 3.7.0; <https://www.python.org/>). All qualitative variables were analyzed using the chi-square test. The threshold for statistical significance was set at $P < 0.05$ (two-tailed). Within the R environment, all necessary packages can be downloaded from <https://cran.r-project.org/web/packages/>.

Results

Baseline characteristics

[Table 1](#) summarizes the baseline information of all patients in the study. The training cohort consisted of 2,263 patients, including 510 GR cases, whereas the validation cohort

TABLE 1 Baseline characteristics of patients.

Characteristics	Total (<i>n</i> = 3,233)	Training cohort (<i>n</i> = 2,263)	Validation cohort (<i>n</i> = 970)	<i>P</i> value
Group (non-GR/GR)	2,504/729	1,753/510	751/219	0.980
Gender (male/female)	1,263/1,970	895/1,368	368/602	0.390
Age (<65 years/≥65 years)	2,305/928	1,604/659	701/269	0.424
BMI (normal/<18.5 or >28)	2,520/713	1,759/504	761/209	0.649
Educational attainment (junior/senior)	1,434/1,799	1,000/1,263	434/536	0.772
Pyloric obstruction (absence/prsence)	2,899/334	2026/237	873/97	0.686
Intestinal obstruction (absence/prsence)	2,993/240	2,094/169	899/71	0.883
Dyspepsia (absence/prsence)	2,903/330	2,038/225	865/105	0.448
History of gastrointestinal surgery (absence/prsence)	2,701/532	1,905/358	796/174	0.372
Systemic disease (absence/prsence)	2,039/1,194	1,420/843	619/351	0.565
Diabetes mellitus (absence/prsence)	2,676/557	1,881/382	795/175	0.423
Thyroid disorders (absence/prsence)	2,978/255	2,084/179	894/76	0.942
Hypoproteinemia (absence/prsence)	2,880/353	2018/245	862/108	0.797
Hypokalemia (absence/prsence)	2,891/342	2,039/224	852/118	0.055
Hyponatremia (absence/prsence)	2,932/301	2,060/203	872/98	0.310
Use of anticholinergic agents (absence/prsence)	2,371/862	1,712/551	659/311	0.356
Use of analgesics (absence/prsence)	2,378/855	1,660/603	718/252	0.694
Use of hypoglycemic agents (absence/prsence)	3,016/217	2,117/146	899/71	0.366
Use of antihistamines (absence/prsence)	2,968/265	2,085/178	883/87	0.295
Use of sedatives (absence/prsence)	2,597/636	1,809/454	788/182	0.395

GR, gastric retention.

comprised 970 patients, with 219 GR cases. No statistically significant differences ($P > 0.05$) were observed between the training and validation cohorts in any characteristics. As shown in Table 2, within the training cohort, significant differences were noted between the non-GR and GR groups regarding age, gender, pyloric obstruction, dyspepsia, history of gastrointestinal surgery, systemic diseases, diabetes, thyroid disorders, hypoalbuminemia, hyponatremia, use of anticholinergic drugs, and use of hypoglycemic agents. In the validation cohort, significant differences were observed in age, BMI, pyloric obstruction, systemic diseases, diabetes mellitus, thyroid disorders, hypoproteinemia, hyponatremia, use of anticholinergic agents, and use of hypoglycemic agents. No statistically significant differences were observed in other characteristics between the two groups ($P > 0.05$).

Clinical characteristics in the SMOTE-enhanced training cohort

Table 3 shows the clinical characteristics of patients in the SMOTE-enhanced training cohort. Notable differences were identified between the two groups regarding BMI, pyloric obstruction, intestinal obstruction, dyspepsia, history of gastrointestinal surgery, systemic disease, diabetes mellitus,

hypoproteinemia, hypokalemia, use of anticholinergic agents, use of hypoglycemic agents, and use of antihistamines ($P < 0.05$). No statistically significant differences were found in other characteristics ($P > 0.05$).

Feature selection and model development

Upon conducting LASSO regression analysis on the SMOTE-enhanced training cohort, the minimum- λ rule retained 19 features, whereas the 1-standard-error rule retained 12 features. Considering model parsimony and clinical interpretability, we selected the 1-standard-error rule with $\lambda = 0.014$ and $\log(\lambda) = -4.243$ (Figure 2). The 12 features retained under this criterion were: use of anticholinergic agents, pyloric obstruction, use of analgesics, systemic disease, history of gastrointestinal surgery, dyspepsia, intestinal obstruction, hypoproteinemia, BMI, use of antihistamines, diabetes mellitus, and hypokalemia. Multicollinearity diagnostics for the 12 predictors showed that all VIF values ranged from 1.043 to 2.294 and were below 5 (Supplementary Table S2). Pairwise correlation analysis demonstrated weak correlations between variables, and no substantial multicollinearity was observed (Supplementary Figure S1). Subsequently, multiple machine learning-based prediction models were constructed using these 12 selected features.

TABLE 2 Comparison of clinical characteristics between two non-GR and GR groups in the training and validation cohorts.

Characteristics	Training cohort			Validation cohort		
	Non-GR (n = 1,753)	GR (n = 510)	P value	Non-GR (n = 751)	GR (n = 219)	P value
Gender (male/female)	716/1,037	179/331	0.019 ^a	286/465	82/137	0.864
Age (<65 years/≥65 years)	1,266/487	338/172	0.009 ^a	557/194	144/75	0.014 ^a
BMI (normal/<18.5 or >28)	1,374/379	385/125	0.167	604/147	157/62	0.006 ^a
Educational attainment (junior/senior)	775/978	225/285	0.971	335/416	99/120	0.875
Pyloric obstruction (absence/prsence)	1,624/129	402/108	<0.001 ^a	697/54	176/43	<0.001 ^a
Intestinal obstruction (absence/prsence)	1,619/134	475/35	0.555	696/55	203/16	0.993
Dyspepsia (absence/prsence)	1,567/186	471/39	0.049 ^a	665/86	200/19	0.245
History of gastrointestinal surgery (absence/ prsence)	1,478/275	405/106	0.046 ^a	638/113	181/38	0.054
Systemic disease (absence/prsence)	1,168/585	252/258	<0.001 ^a	511/240	108/111	<0.001 ^a
Diabetes mellitus (absence/prsence)	1,500/253	381/129	<0.001 ^a	637/114	158/61	<0.001 ^a
Thyroid disorders (absence/prsence)	1,633/120	451/59	<0.001 ^a	700/51	194/25	0.025 ^a
Hypoproteinemia (absence/prsence)	1,602/151	416/94	<0.001 ^a	678/73	184/35	0.010 ^a
Hypokalemia (absence/prsence)	1,586/167	453/57	0.272	663/88	189/30	0.430
Hyponatremia (absence/prsence)	1,615/138	445/65	<0.001 ^a	693/58	179/40	<0.001 ^a
Use of anticholinergic agents (absence/prsence)	1,295/458	354/156	0.011 ^a	568/183	144/75	<0.001 ^a
Use of analgesics (absence/prsence)	1,298/455	362/148	0.168	555/196	163/56	0.875
Use of hypoglycemic agents (absence/prsence)	1,680/73	437/73	<0.001 ^a	714/37	185/34	<0.001 ^a
Use of antihistamines (absence/prsence)	1,624/129	461/49	0.097	691/60	192/27	0.048 ^a
Use of sedatives (absence/prsence)	1,397/356	412/98	0.588	605/146	183/36	0.317

GR, gastric retention.
^aStatistically significant at $P < 0.05$.

Performance of different models

The ROC curve illustrates the performance of each model in the SMOTE-enhanced training cohort and validation cohort, as depicted in Figure 3. Figure 4 and Supplementary Table S3 detail the performance of various prediction models in the SMOTE-enhanced training and validation cohorts, whereas Supplementary Figures S2, S3 depict the corresponding confusion matrices of these models in the SMOTE-enhanced training and validation cohorts. The results indicate that the nine machine learning models achieved AUC ranges of 0.687–0.806 and 0.563–0.710 in the SMOTE-enhanced training cohort and validation cohort, respectively. The XGBoost model demonstrated the best performance in both cohorts. Comparisons of AUC among different prediction models are presented in Supplementary Figure S4. The XGBoost model achieved an average AUC of 0.766 ± 0.019 and an average accuracy of 0.691 ± 0.019 (Supplementary Figure S5) when 10-fold cross-validation was employed in the SMOTE-enhanced training cohort. Calibration curves demonstrated good agreement between predicted and observed probabilities in both cohorts (Supplementary Figure S6), and DCA indicated that this model provided a superior net benefit over the others across a range of threshold probabilities (Figure 5). Threshold-specific performance analysis further revealed that as the probability

cutoff increased, sensitivity progressively decreased while specificity improved (Supplementary Figure S7). Based on the Youden index, the optimal probability threshold was determined to be 0.418, yielding a sensitivity of 77.6%, specificity of 53.3%, PPV of 32.6%, and NPV of 89.1% in the validation cohort. Notably, this Youden-optimized cutoff leads to over 45% of GR-negative patients being misclassified as high-risk individuals, which may cause unnecessary prolonged fasting and endoscopy delays. We further calculated three representative alternative cutoffs for differentiated clinical use: a high-sensitivity threshold of 0.278 (sensitivity = 85.4%, specificity = 37.0%), a balanced threshold of 0.510 (sensitivity = 60.3%, specificity = 69.0%), and a high-specificity threshold of 0.624 (sensitivity = 50.7%, specificity = 76.0%). Full numerical results for all candidate thresholds are summarized in Supplementary Table S4.

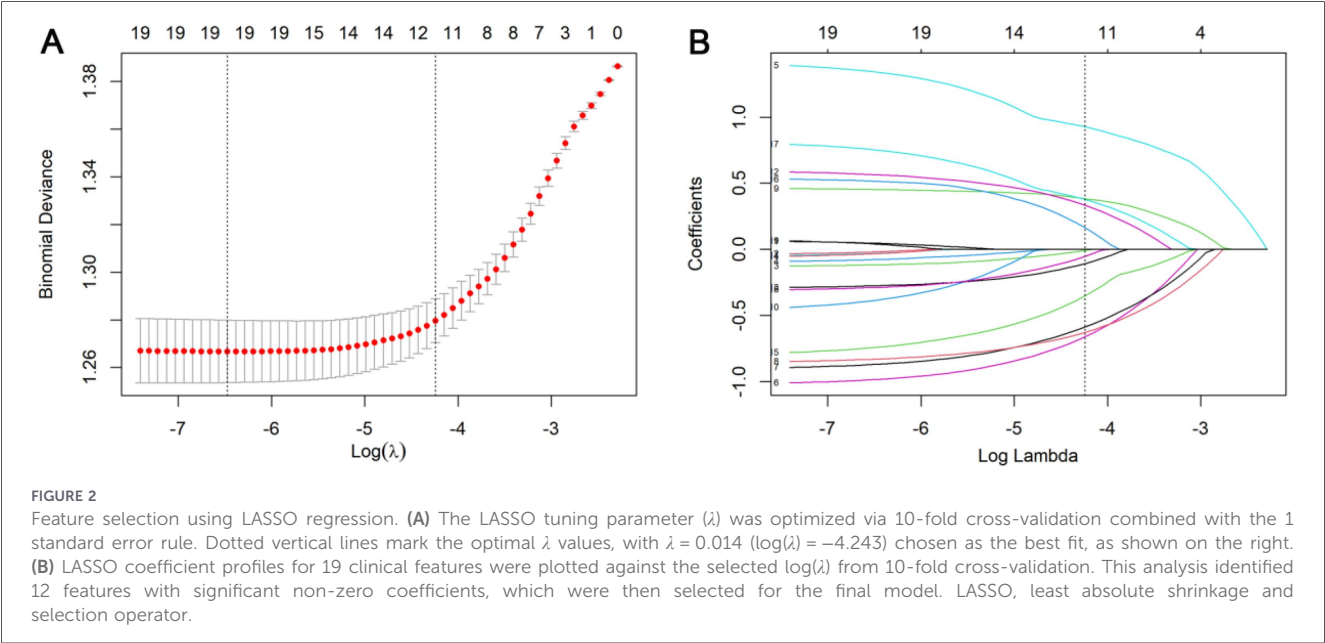
Interpretation of the XGBoost model

Visualization analysis of the XGBoost model was conducted using the SHAP algorithm. The bar chart indicates that the most influential feature on the model is the use of anticholinergic agents, followed by pyloric obstruction, use of analgesics, systemic disease, and history of gastrointestinal surgery (Figure 6). The beeswarm plot demonstrates that the use

TABLE 3 Comparison of patients' clinical characteristics between non-GR and GR in the SMOTE-enhanced training cohort.

Characteristics	Non-GR (n = 1,753)	GR (n = 1,753)	P value
Gender (male/female)	716/1,037	678/1,075	0.190
Age (<65 years/≥65 years)	1,266/487	1,251/502	0.573
BMI (normal/<18.5 or >28)	1,374/379	1,446/307	0.002 ^a
Educational attainment (junior/senior)	775/978	818/935	0.145
Pyloric obstruction (absence/prsence)	1,624/129	1,374/379	<0.001 ^a
Intestinal obstruction (absence/prsence)	1,619/134	1,696/57	<0.001 ^a
Dyspepsia (absence/prsence)	1,567/186	1,676/77	<0.001 ^a
History of gastrointestinal surgery (absence/prsence)	1,478/275	1,332/421	<0.001 ^a
Systemic disease (absence/prsence)	1,168/585	953/800	<0.001 ^a
Diabetes mellitus (absence/prsence)	1,500/253	1,318/435	<0.001 ^a
Thyroid disorders (absence/prsence)	1,633/120	1,640/113	0.635
Hypoproteinemia (absence/prsence)	1,602/151	1,532/221	<0.001 ^a
Hypokalemia (absence/prsence)	1,586/167	1,635/118	0.002 ^a
Hyponatremia (absence/prsence)	1,615/138	1,603/150	0.460
Use of anticholinergic agents (absence/prsence)	1,295/458	1,116/637	<0.001 ^a
Use of analgesics (absence/prsence)	1,298/455	1,311/442	0.615
Use of hypoglycemic agents (absence/prsence)	1,680/73	1,552/201	<0.001 ^a
Use of antihistamines (absence/prsence)	1,624/129	1,661/92	0.010 ^a
Use of sedatives (absence/prsence)	1,397/356	1,439/314	0.071

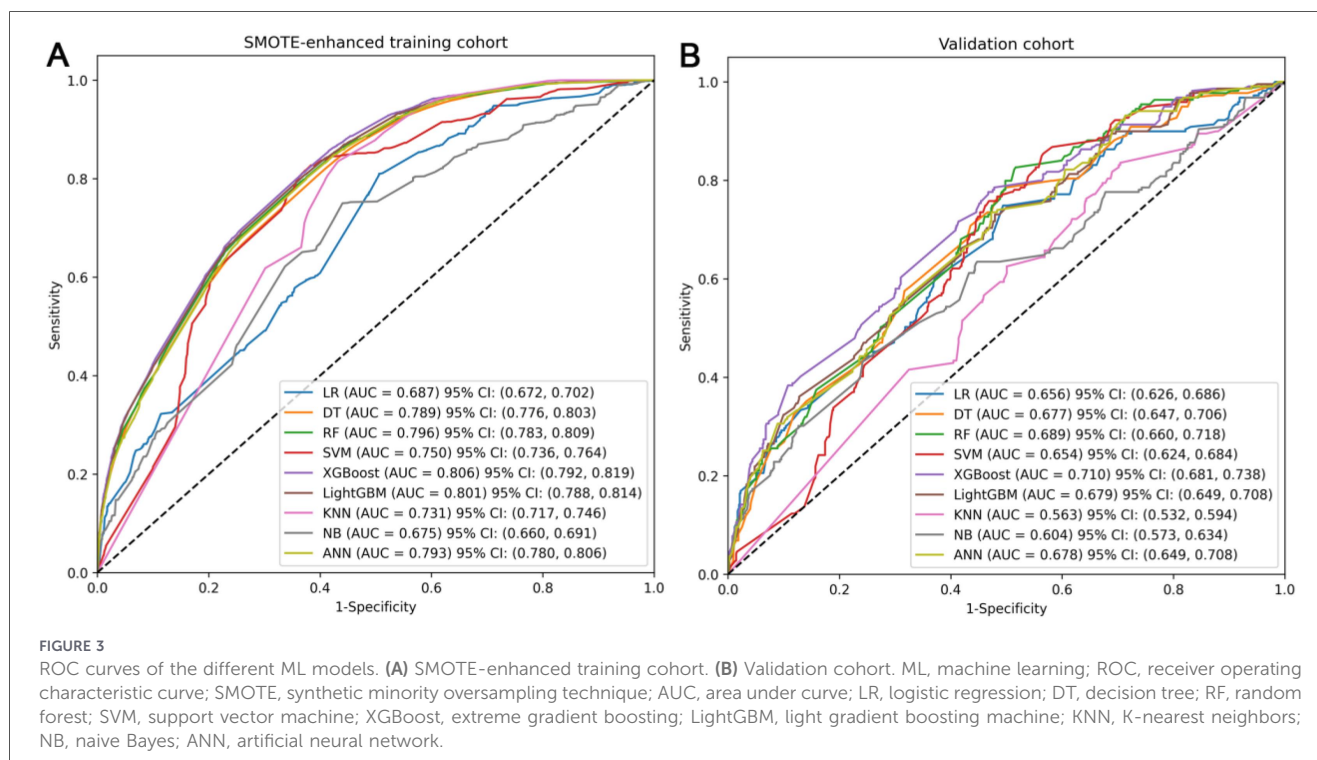
GR, gastric retention; SMOTE, synthetic minority over-sampling technique.
^aStatistically significant at $P < 0.05$.



of anticholinergic agents, pyloric obstruction, use of analgesics, systemic disease, and history of gastrointestinal surgery are associated with increased GR risk (Figure 7). Figure 8 displays the force plots and waterfall plots for two representative patients (non-GR and GR) from the validation set.

Discussion

This study represents the large-scale dataset-based study to develop and validate an interpretable ML model for predicting GR risk before gastroscopy. Our analysis of 3,233 patients



identified 12 key clinical predictors and demonstrated that the XGBoost algorithm outperformed other ML models, achieving an AUC of 0.806 in the SMOTE-enhanced training cohort and 0.710 in the validation cohort.

The performance of the XGBoost model is consistent with its established strengths in handling complex, non-linear interactions within imbalanced medical datasets. Its AUC of 0.710 in the validation cohort reflects an expected decrease from training performance and falls within the moderate discrimination range. The model's calibration curve showed good agreement between predicted probabilities and observed outcomes, and the DCA curve confirmed its net clinical benefit across a range of threshold probabilities, supporting its potential utility as an auxiliary screening tool, though it should not replace comprehensive clinical judgment. At the optimal probability cutoff of 0.418 (determined by maximizing the Youden index), the model achieved a sensitivity of 77.6% and an NPV of 89.1%, indicating that it could safely rule out GR in nearly 90% of low-risk patients, while the moderate PPV of 32.6% reflects the expected impact of the low disease prevalence (22.6%) on positive predictions. In the validation cohort, XGBoost achieved an AUC of 0.710, compared with 0.656 for logistic regression. XGBoost also showed higher sensitivity and NPV, whereas its accuracy and specificity were not higher than those of logistic regression. Additionally, clinical interpretability via SHAP analysis provides a potentially useful adjunct to traditional risk stratification approaches (12).

We further elaborate on the trade-off between sensitivity and specificity when selecting model thresholds in clinical practice. The threshold of 0.418 derived from the Youden index prioritizes overall statistical performance yet suffers from low specificity (53.3%), producing a high false-positive rate. The balanced cutoff of 0.510 is more suitable for routine gastroscopy

scenarios, as it markedly reduces false positive cases while retaining acceptable predictive ability, with balanced sensitivity (60.3%) and specificity (69.0%). For sedated endoscopy where preventing aspiration is the top priority, clinicians can adopt the low threshold of 0.278 to maximize sensitivity (85.4%) and avoid missed GR cases, even though this will greatly increase the number of patients requiring extended fasting. Conversely, the high threshold of 0.624 can be applied in unsedated outpatient endoscopy to lift specificity to 76.0%, easing hospital scheduling burdens at the cost of partially reduced sensitivity. Clinicians can flexibly adjust the probability cutoff according to anesthesia mode, patient conditions and institutional operational demands to balance clinical safety and work efficiency.

We found that the use of anticholinergic agents emerged as the strongest predictor of GR. This is mechanistically plausible, as anticholinergics directly inhibit gastric smooth muscle contraction and delay gastric emptying. Similarly, the use of analgesics, antihistamines, and hypoglycemic agents were also significant contributors. These findings emphasize the crucial role of medication review in GR risk assessment. Pyloric obstruction and a history of gastrointestinal surgery were among the top predictors. Obstruction physically impedes emptying, while surgery can disrupt neural (vagal) or anatomical pathways essential for normal motility (18, 19). A history of gastrointestinal surgery may cause GR by altering normal anatomy and damaging the vagus nerve, leading to abnormal peristalsis and food residue (2, 20–23). However, intestinal obstruction has a lesser impact on GR, possibly because the condition is more severe, and gastroscopy is not a necessary examination for such patients.

The presence of systemic diseases, diabetes mellitus, hypoproteinemia, and hypokalemia were significant predictors. Hypoproteinemia may reflect malnutrition or systemic

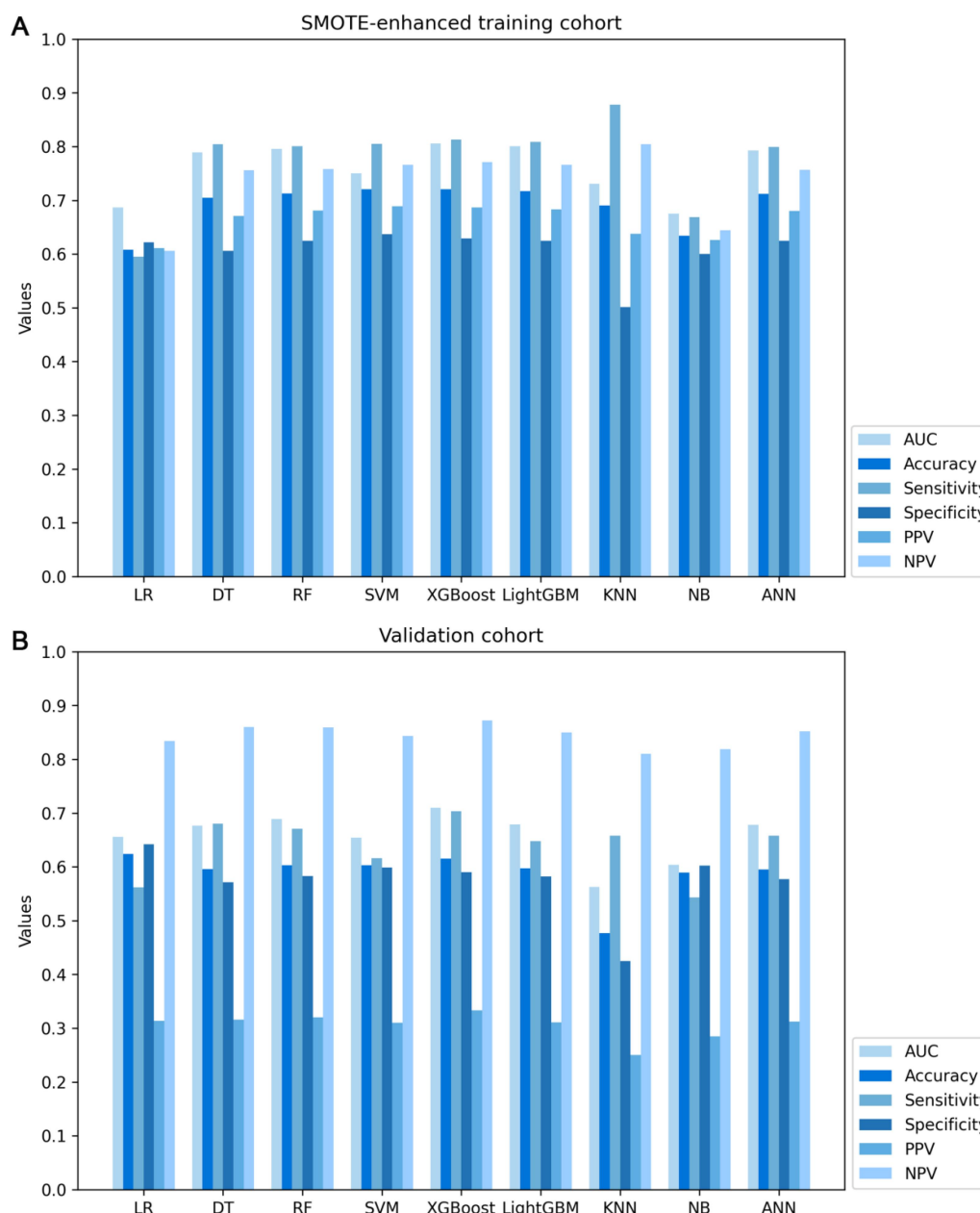


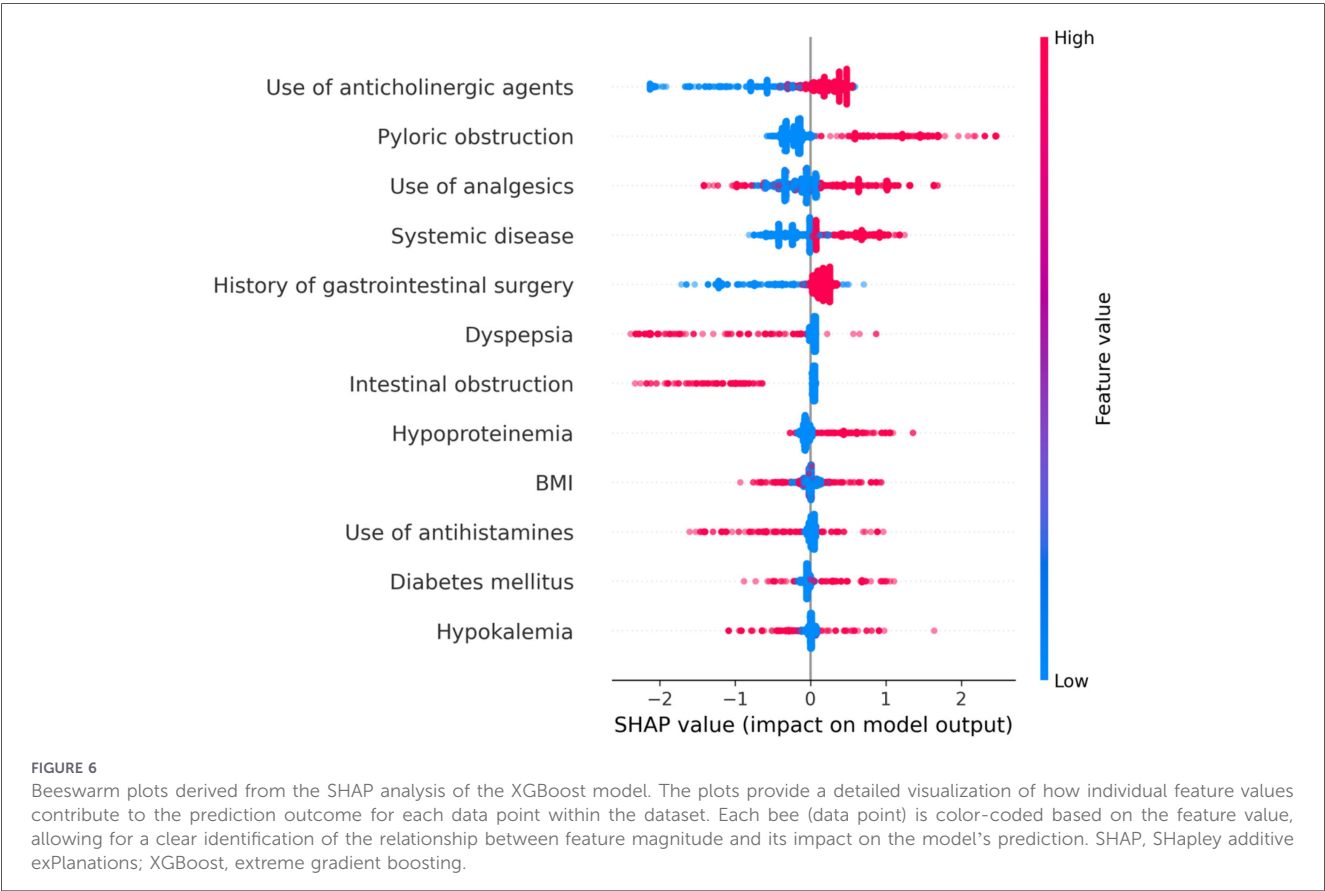
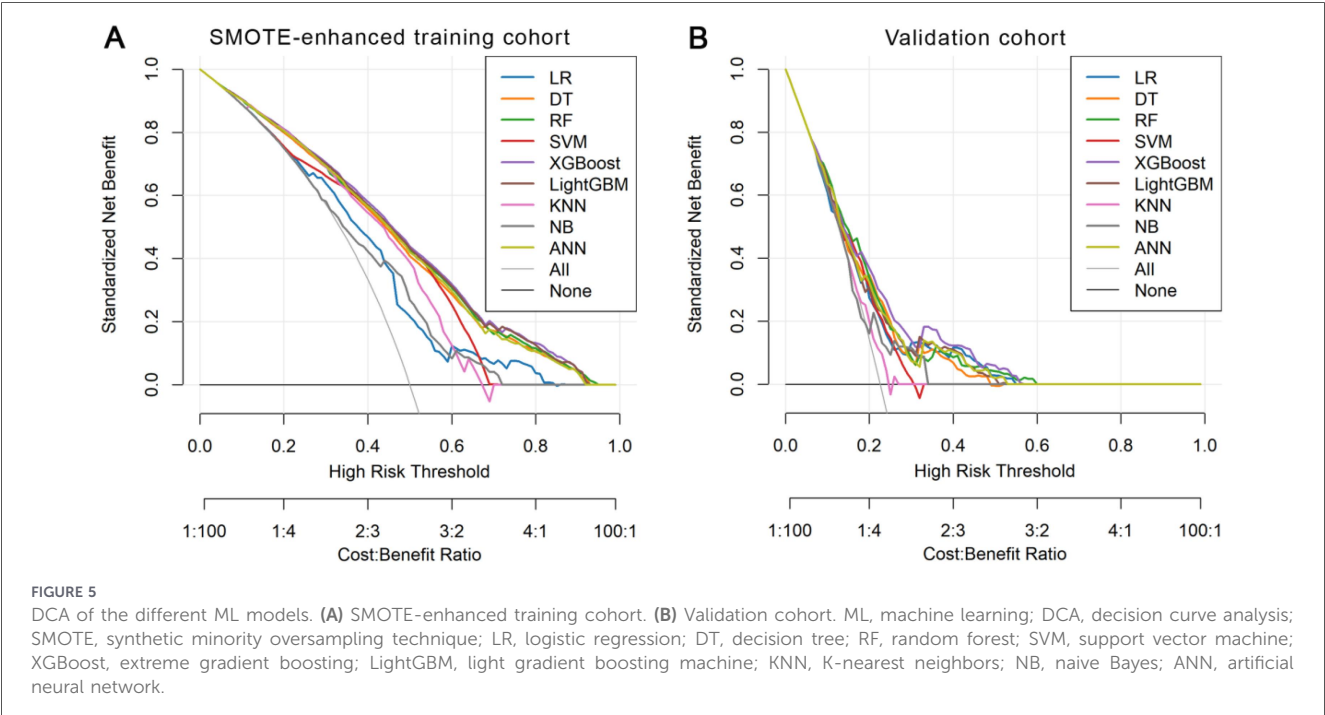
FIGURE 4

Performances of the different ML models. (A) SMOTE-enhanced training cohort. (B) Validation cohort. ML, machine learning; SMOTE, synthetic minority oversampling technique; AUC, area under receiver operating characteristics; CI, confidence interval; Acc, accuracy; Sen, sensitivity; Spe, specificity; PPV, positive predictive value; NPV, negative predictive value; LR, logistic regression; DT, decision tree; RF, random forest; SVM, support vector machine; XGBoost, extreme gradient boosting; LightGBM, light gradient boosting machine; KNN, K-nearest neighbors; NB, naive Bayes; ANN, artificial neural network. All metrics were calculated at a 0.5 threshold for model comparison.

inflammation affecting motility, while hypokalemia can directly impair smooth muscle function (24). Current evidence regarding BMI and gastric motility remains conflicting. In our study, BMI <18.5 or BMI >28 showed no significant association with gastric retention risk when compared to normal BMI controls. While some studies suggest that low BMI may contribute to gastric retention, others indicate that elevated BMI accelerates gastric emptying rates (25–28). Results from many studies in patients with diabetes indicate that diabetes is a risk factor for gastroparesis (29, 30). However, in our study, diabetes was not among the top five factors affecting GR, possibly

because the diabetic patients received timely medication intervention and were advised to extend fasting time.

Several previous studies have evaluated gastric food retention, residual gastric content, or gastroparesis prediction in specific clinical contexts. Coleski et al. and Phillips et al. identified important clinical factors associated with endoscopic gastric food retention or high-risk residual gastric content, but these studies mainly focused on risk-factor identification rather than developing an interpretable machine-learning risk stratification tool for routine pre-gastroscopy use (1, 31). Jia et al. developed a logistic regression-based model for preoperative gastric



retention before ERCP, with good discrimination in a small validation cohort; however, ERCP represents a more selected therapeutic endoscopy population, and the sample size was substantially smaller than that of the present study (32). Other prediction models have focused on postsurgical gastroparesis syndrome after gastrectomy or complete mesocolic excision, or on gastroparesis in chronic pancreatitis. Although some of these disease- or surgery-specific models reported higher AUC values,

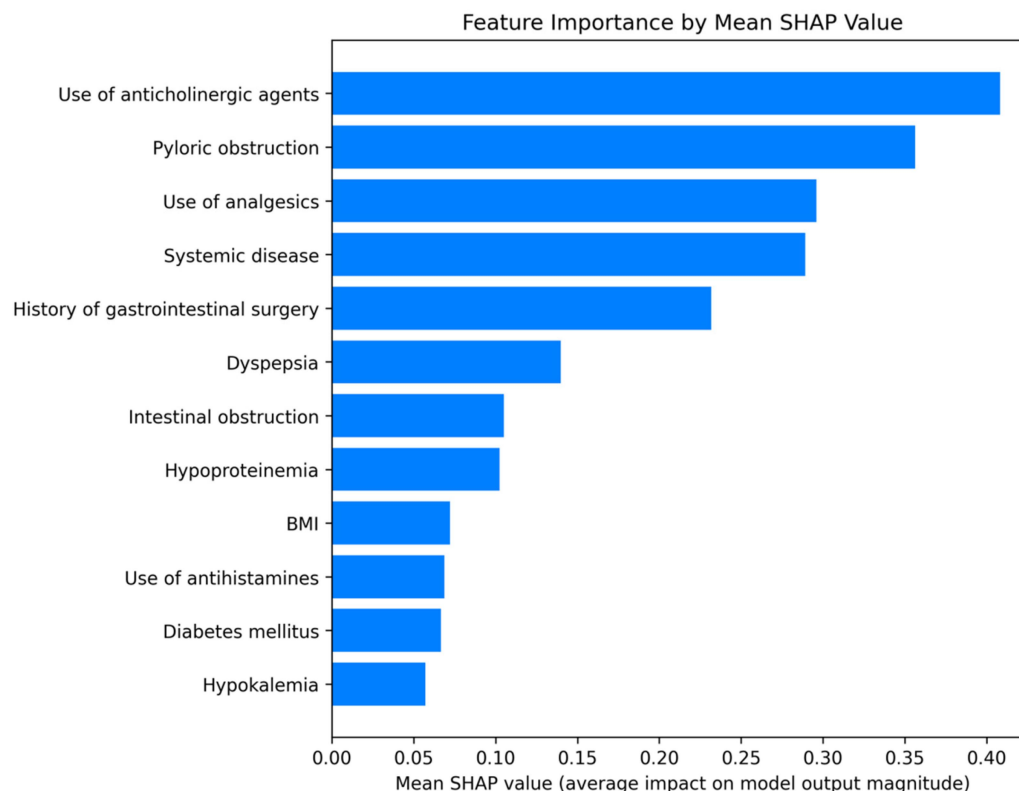


FIGURE 7

Bar plots from the SHAP analysis of the XGBoost model. The length of each bar represents the overall contribution of a specific feature to the model's predictions across the entire dataset. SHAP, SHapley additive exPlanations; XGBoost, extreme gradient boosting.

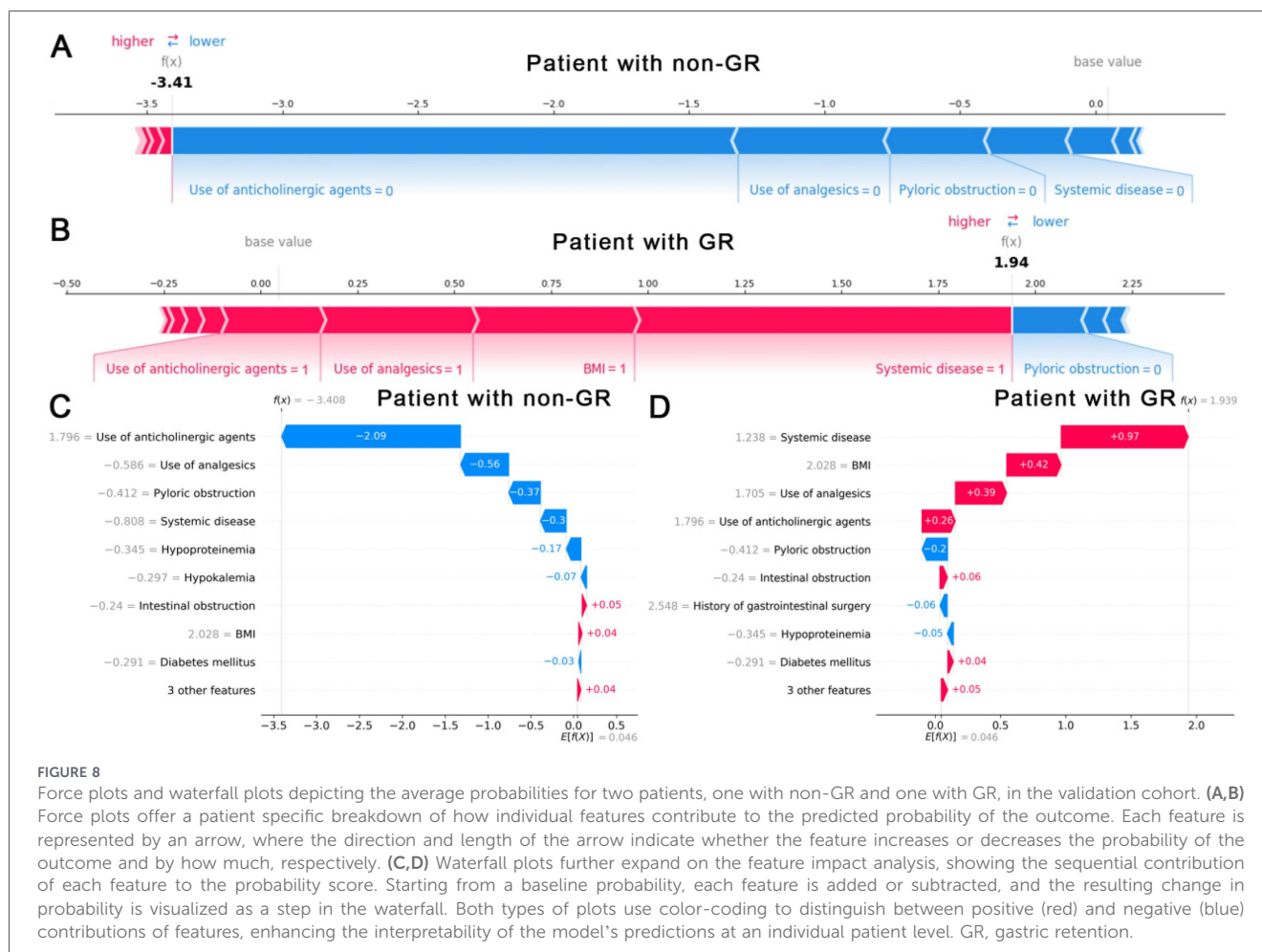
their target populations, endpoints, and predictors differ substantially from those required for routine gastroscopy screening.

Compared with these prior studies, the present model has several practical advantages for the specific task of predicting gastric retention before gastroscopy. First, it was developed in a larger routine gastroscopy cohort and used routinely available pre-procedural variables, making it more suitable for integration into standard endoscopy workflows. Second, using the same LASSO-selected predictors, we evaluated nine algorithms (with LR as baseline). In the validation cohort, XGBoost showed higher discrimination than LR (AUC: 0.710 vs. 0.656), along with superior sensitivity and NPV. Third, the model was evaluated not only by ROC analysis but also by calibration, decision curve analysis, confusion matrices, and SHAP-based interpretation. Therefore, although external validation is still required, the proposed XGBoost model offers a useful reference that is more comprehensive and clinically interpretable for pre-gastroscopy GR risk stratification than prior risk-factor studies or models developed for other disease- or procedure-specific populations.

The integration of machine learning into clinical practice continues to expand, offering robust tools for developing predictive models. These models enable precision medicine through individualized risk stratification and treatment optimization, thereby enhancing patient outcomes (7, 13, 15). By integrating readily available clinical data (demographics, history, medications, basic labs), this model may enable early

identification of high-risk patients for GR. High-risk patients could undergo extended fasting protocols, receive prokinetic agents, or have their gastroscopy scheduled as the first procedure of the day. Identifying low-risk patients could potentially allow for more streamlined fasting protocols, improving patient comfort and operational efficiency. Ultimately, this stratification aims to significantly reduce the incidence of aspiration pneumonia and asphyxiation. Currently, challenges such as the integration of electronic health records, automated extraction of drug and laboratory variables, cross-institutional model calibration, and physician training require urgent resolution to optimize endoscopic examination scheduling and anesthesia assessment workflows.

This study has several limitations. Firstly, as a single-center retrospective study, it may have selection bias. This study lacks external independent data validation; therefore, prospective, multi-center validation studies are essential to confirm the model's generalizability and real-world clinical impact. Secondly, factors such as glycated hemoglobin level and gastric emptying scintigraphy were not included in the study due to limited conditions. The inclusion of more detailed clinical data is expected to improve the predictive performance of the model. Third, SMOTE oversampling was performed prior to LASSO feature selection. Artificially synthesized minority samples cannot fully mimic the real distribution of patients with gastric retention, and this analytical pipeline may introduce some degree of optimism during feature selection and model training. Although we conducted 10-fold cross-validation within the



SMOTE-enhanced training cohort to assess model stability on the balanced dataset, this cross-validation cannot completely eliminate the bias introduced by pre-analysis oversampling. Future investigations on independent naturally imbalanced cohorts are warranted to further verify the robustness of the identified predictors. Lastly, while SMOTE effectively addressed class imbalance for model training, the synthetic samples may not perfectly reflect the true underlying distribution of the minority GR class. Additionally, the moderate AUC of 0.710 indicates that the current model lacks the precision required for standalone clinical decision-making. Nevertheless, this level of discrimination is comparable to that of existing risk stratification tools for similar clinical scenarios, supporting its potential as an auxiliary screening aid, though further refinement through the inclusion of more comprehensive objective indicators is needed before broader clinical implementation.

Conclusions

We developed an interpretable XGBoost model using 12 routinely collected clinical features to predict GR risk before gastroscopy. The model showed moderate discrimination with good calibration. SHAP analysis identified anticholinergic use, pyloric obstruction, gastrointestinal surgery history, analgesic use, and systemic disease as key contributing features. The

model enables early identification of high-risk patients to guide pre-procedural interventions, but should be viewed as an auxiliary screening aid rather than a definitive diagnostic tool. Future prospective external validation is warranted.

Data availability statement

The original contributions presented in the study are included in the article/[Supplementary Material](#), further inquiries can be directed to the corresponding author/s.

Ethics statement

The studies involving humans were approved by The Institutional Review Board of the First Affiliated Hospital of Xiamen University (XMY-2025KY173). The studies were conducted in accordance with the local legislation and institutional requirements. The ethics committee/institutional review board waived the requirement of written informed consent for participation from the participants or the participants' legal guardians/next of kin due to the retrospective nature of the study.

Author contributions

YD: Conceptualization, Funding acquisition, Writing – original draft, Writing – review & editing. HS: Conceptualization, Data curation, Methodology, Writing – review & editing. HZ: Conceptualization, Data curation, Writing – review & editing. JC: Conceptualization, Methodology, Writing – review & editing. GX: Data curation, Writing – review & editing. HW: Conceptualization, Writing – review & editing.

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Conflict of interest

The author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

References

- Coleski R, Baker JR, Hasler WL. Endoscopic gastric food retention in relation to scintigraphic gastric emptying delays and clinical factors. *Dig Dis Sci.* (2016) 61(9):2593–601. doi: 10.1007/s10620-016-4173-7
- Takahashi R, Ohashi M, Hiki N, Makuuchi R, Ida S, Kumagai K, et al. Risk factors and prognosis of gastric stasis, a crucial problem after laparoscopic pylorus-preserving gastrectomy for early middle-third gastric cancer. *Gastric Cancer.* (2020) 23(4):707–15. doi: 10.1007/s10120-019-01037-4
- Takayama H, Toyonaga T, Yoshizaki T, Abe H, Nakai T, Ueda C, et al. Characteristics of gastric stasis due to deformation after endoscopic submucosal dissection in the lower part of the stomach. *Digestion.* (2023) 104(4):320–7. doi: 10.1159/000529316
- Kaji S, Makuuchi R, Irino T, Tanizawa Y, Bando E, Kawamura T, et al. Preventive effect on delayed gastric emptying of preserving the infra-pyloric vein in laparoscopic pylorus-preserving gastrectomy for early gastric cancer. *Surg Endosc.* (2020) 34(9):3853–60. doi: 10.1007/s00464-019-01751-9
- Hasler WL. Gastroparesis: pathogenesis, diagnosis and management. *Nat Rev Gastroenterol Hepatol.* (2011) 8(8):438–53. doi: 10.1038/nrgastro.2011.116
- Bharucha AE, Kudva YC, Prichard DO. Diabetic gastroparesis. *Endocr Rev.* (2019) 40(5):1318–52. doi: 10.1210/er.2018-00161
- Emegano DI, Duwa BB, Usman AG, Ahmad H, Ohsahin DU, Askar S. A comparative study on TB incidence and HIVTB coinfection using machine learning models on WHO global TB dataset. *Sci Rep.* (2025) 15(1):13690. doi: 10.1038/s41598-025-94378-3
- Sharaha U, Rodriguez-Diaz E, Sagi O, Riesenberger K, Lapidot I, Segal Y, et al. Detection of extended-Spectrum β -lactamase-producing *Escherichia coli* using infrared microscopy and machine-learning algorithms. *Anal Chem.* (2019) 91(3):2525–30. doi: 10.1021/acs.analchem.8b05497
- John L, Dcunha L, Ahmed M, Thomas SD, Raju R, Jayanandan A. A deep learning and molecular modeling approach to repurposing cangrelor as a potential inhibitor of nipah virus. *Sci Rep.* (2025) 15(1):16440. doi: 10.1038/s41598-025-00024-3
- Huang Q, Zou X, Lian Z, Zhou X, Han X, Luo Y, et al. Predicting cardiovascular outcomes in Chinese patients with type 2 diabetes by combining risk factor trajectories and machine learning algorithm: a cohort study. *Cardiovasc Diabetol.* (2025) 24(1):61. doi: 10.1186/s12933-025-02611-0
- Ogami C, Tsuji Y, Seki H, Kawano H, To H, Matsumoto Y, et al. An artificial neural network-pharmacokinetic model and its interpretation using shapley

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Supplementary material

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- additive explanations. *CPT Pharmacometrics Syst Pharmacol.* (2021) 10(7):760–8. doi: 10.1002/psp4.12643
- Cao S, Hu Y. Creating machine learning models that interpretably link systemic inflammatory index, sex steroid hormones, and dietary antioxidants to identify gout using the SHAP (SHapley additive eXplanations) method. *Front Immunol.* (2024) 15:1367340. doi: 10.3389/fimmu.2024.1367340
- McFall GP, Bohn L, Gee M, Drouin SM, Fah H, Han W, et al. Identifying key multi-modal predictors of incipient dementia in Parkinson's disease: a machine learning analysis and tree SHAP interpretation. *Front Aging Neurosci.* (2023) 15:1124232. doi: 10.3389/fnagi.2023.1124232
- Rahman W, Lee S, Islam MS, Antony VN, Ratnu H, Ali MR, et al. Detecting Parkinson disease using a web-based speech task: observational study. *J Med Internet Res.* (2021) 23(10):e26305. doi: 10.2196/26305
- Yu J, Peng X, Zhou R, Zhu T, Hao X. Development and validation of an interpretable machine learning model to predict major adverse cardiovascular events after noncardiac surgery in geriatric patients: a prospective study. *Int J Surg.* (2025) 111(2):1939–49. doi: 10.1097/S19.0000000000002203
- Talley NJ, Locke GR, Lahr BD, Zinsmeister AR, Tougas G, Ligozio G, et al. Functional dyspepsia, delayed gastric emptying, and impaired quality of life. *Gut.* (2006) 55(7):933–9. doi: 10.1136/gut.2005.078634
- Chawla NV, Bowyer KW, Hall LO, Kegelmeyer WP. SMOTE: synthetic minority over-sampling technique. *J Artif Intell Res.* (2002) 10(7):321. doi: 10.1613/jair.953
- Gurvits GE, Tan A, Volkov D. Video capsule endoscopy and CT enterography in diagnosing adult hypertrophic pyloric stenosis. *World J Gastroenterol.* (2013) 19(37):6292–5. doi: 10.3748/wjg.v19.i37.6292
- Camilleri M. Functional dyspepsia and gastroparesis. *Dig Dis.* (2016) 34(5):491–9. doi: 10.1159/000445226
- Yu Z, Zhao X, Qiu S, Liu N, Li P, Zhou S. Risk factor analysis of gastroparesis syndrome in 2652 patients with radical distal gastrectomy. *J Gastrointest Surg.* (2023) 27(8):1568–77. doi: 10.1007/s11605-022-05538-z
- Meng H, Zhou D, Jiang X, Ding W, Lu L. Incidence and risk factors for postsurgical gastroparesis syndrome after laparoscopic and open radical gastrectomy. *World J Surg Oncol.* (2013) 11:144. doi: 10.1186/1477-7819-11-144
- Qu H, Sun GR, Zhou SQ, He QS. Clinical risk factors of delayed gastric emptying in patients after pancreaticoduodenectomy: a systematic review and meta-analysis. *Eur J Surg Oncol.* (2013) 39(3):213–23. doi: 10.1016/j.ejso.2012.12.010

23. Yoshizaki T, Obata D, Aoki Y, Okamoto N, Hashimura H, Kano C, et al. Endoscopic submucosal dissection for early gastric cancer on the lesser curvature in upper third of the stomach is a risk factor for postoperative delayed gastric emptying. *Surg Endosc.* (2018) 32(8):3622–9. doi: 10.1007/s00464-018-6091-6
24. Türck G, Leonhard-Marek S. Potassium and insulin affect the contractility of abomasal smooth muscle. *J Dairy Sci.* (2010) 93(8):3561–8. doi: 10.3168/jds.2009-2968
25. Limketkai BN, LeBrett W, Lin L, Shah ND. Nutritional approaches for gastroparesis. *Lancet Gastroenterol Hepatol.* (2020) 5(11):1017–26. doi: 10.1016/S2468-1253(20)30078-9
26. Gonzalez-Izundegui D, Campos A, Calderon G, Ricardo-Silgado ML, Cifuentes L, Decker PA, et al. Association of gastric emptying with postprandial appetite and satiety sensations in obesity. *Obesity (Silver Spring).* (2021) 29(9):1497–507. doi: 10.1002/oby.23204
27. Buchholz V, Berkenstadt H, Goitein D, Dickman R, Bernstine H, Rubin M. Gastric emptying is not prolonged in obese patients. *Surg Obes Relat Dis.* (2013) 9(5):714–7. doi: 10.1016/j.soard.2012.03.008
28. Lu CX, An XX, Yu Y, Jiao LR, Canarutto D, Li GF, et al. Pooled analysis of gastric emptying in patients with obesity: implications for oral absorption projection. *Clin Ther.* (2021) 43(10):1768–88. doi: 10.1016/j.clinthera.2021.08.006
29. Camilleri M, Bharucha AE, Farrugia G. Epidemiology, mechanisms, and management of diabetic gastroparesis. *Clin Gastroenterol Hepatol.* (2011) 9(1):5–12. doi: 10.1016/j.cgh.2010.09.022
30. Choung RS, Locke RG, Schleck CD, Zinsmeister AR, Melton JL, Talley NJ. Risk of gastroparesis in subjects with type 1 and 2 diabetes in the general population. *Am J Gastroenterol.* (2012) 107(1):82–8. doi: 10.1038/ajg.2011.310
31. Phillips S, Liang SS, Formaz-Preston A, Stewart PA. High-risk residual gastric content in fasted patients undergoing gastrointestinal endoscopy: a prospective cohort study of prevalence and predictors. *Anaesth Intensive Care.* (2015) 43(6):728–33. doi: 10.1177/0310057X1504300610
32. Jia Y, Wu HJ, Li T, Liu JB, Fang L, Liu ZM. Establishment of predictive models and determinants of preoperative gastric retention in endoscopic retrograde cholangiopancreatography. *World J Gastrointest Surg.* (2024) 16(8):2574–82. doi: 10.4240/wjgs.v16.i8.2574